

CE 4011 Static + Modal Desktop MVP Verification

Appendix

Final Documentation Package

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1 Purpose and Scope Boundary

This appendix records detailed validation evidence for the submitted desktop MVP. The final desktop analysis scope is Static Analysis and Modal Analysis only. RSA and THA may remain in the repository as deferred/future-extension backend work, but they are not part of the submitted desktop verification claim.

The main report summarizes the validation baseline. This appendix provides the longer benchmark descriptions, acceptance criteria, coverage mapping, and comparison tables. Values from the existing `data/` result exports are inserted where the exported quantity is explicit.

2 Verification Method

The verification strategy combines reference benchmarks and automated tests. Static benchmark models are loaded from XML, solved through the same DOF/assembly/solver/post-processing path used by the desktop application, and compared with SAP2000 text-output references. Modal benchmark models are loaded from XML, assembled dynamically, reduced or condensed where needed, and compared with a flexible-beam modal reference solution.

Each benchmark row reports one scalar comparison. Pass/fail is based on the same tolerances used by the automated benchmark tests, including the absolute tolerance floor used for near-zero displacement and force components.

Table 1: Verification evidence map

Evidence area	Input/reference	Quantities checked	Acceptance basis
Static frame benchmark	Frame XML; SAP2000 frame text output	Nodal displacements, support reactions, member-end forces, global/reduced matrices retained for inspection	Regression comparison against parsed SAP2000 tables, with relative tolerances used by the automated benchmark tests

Evidence area	Input/reference	Quantities checked	Acceptance basis
Settlement benchmark	Settlement XML; SAP2000 settlement text output	Prescribed support settlement response, nodal displacements, reactions, member forces, SAP2000-display mapping	SAP2000 comparison plus display-mapping tolerance used by desktop/static tests
Temperature benchmark	Temperature XML; SAP2000 temperature text output	Thermal fixed-end effects, inclined truss transformations, displacements, reactions, member-end forces	Strict thermal benchmark tolerances for displacement, force, and moment components
Modal flexible-beam benchmark	Modal XML; flexible-beam modal reference text output	Condensed stiffness, eigenvalues, circular frequencies, periods, mode shapes, mass-normalized modal properties where available	Absolute tolerances used by modal regression tests
Desktop command paths	Headless desktop test doubles and temporary files	XML save/export, XML parse-back, Static table TXT export, Modal table CSV export, Static plot PNG export	Final P0 desktop workflow tests

The retained benchmark inputs are `data/test-frame.xml`, `data/test-settlement.xml`, `data/test-temperature.xml`, and `data/test-modal-flex-beam.xml`. The corresponding reference outputs are stored under `sap2000_solutions/` and `modal_solution/`. The final documentation export evidence files are `data/test_frame_static_results_table.txt` and `data/modal_results_table.csv`.

3 Detailed Theory Notes

This section retains the longer equations and implementation notes removed from the main report to keep the submitted main body within the CE 4011 page limit.

For a two-dimensional frame member with local axial and flexural behavior, the local stiffness contribution has the standard direct-stiffness form

$$k_\ell = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}.$$

Truss members use the axial subspace of this relationship. Each element contribution is transformed into global coordinates, assembled into the global stiffness matrix, and partitioned into free and restrained degrees of freedom. With restrained displacement vector u_r , the solved static system is

$$K_{ff}u_f = F_f - K_{fr}u_r, \quad R = Ku - F.$$

Element end forces are recovered by returning the solved global displacement vector to local member coordinates and adding equivalent nodal-load effects where member loads, settlement, or temperature loads apply. The same recovered local-force data supports axial-force, shear-force, and bending-moment diagram sampling.

The modal workflow solves the generalized eigenproblem

$$K_d \phi_i = \lambda_i M_d \phi_i, \quad \omega_i = \sqrt{\lambda_i}, \quad T_i = \frac{2\pi}{\omega_i},$$

on the active dynamic degrees of freedom. Massless DOFs that are stiffness-coupled to retained inertial DOFs are statically condensed rather than deleted:

$$K_c = K_{aa} - K_{am} K_{mm}^{-1} K_{ma}.$$

Direct deletion is reserved for disconnected zero-mass DOFs. Mode shapes are normalized for display and for modal-property reporting. Modal mass, participation factor, effective modal mass, and participation ratio are computed from the retained mass matrix and the selected excitation direction.

4 Desktop Screenshot Evidence

The main report keeps one UI/workflow figure and one result/visualization figure. The appendix-local screenshots below capture the model setup, validation menu, property editing, and export controls used during the desktop workflow.

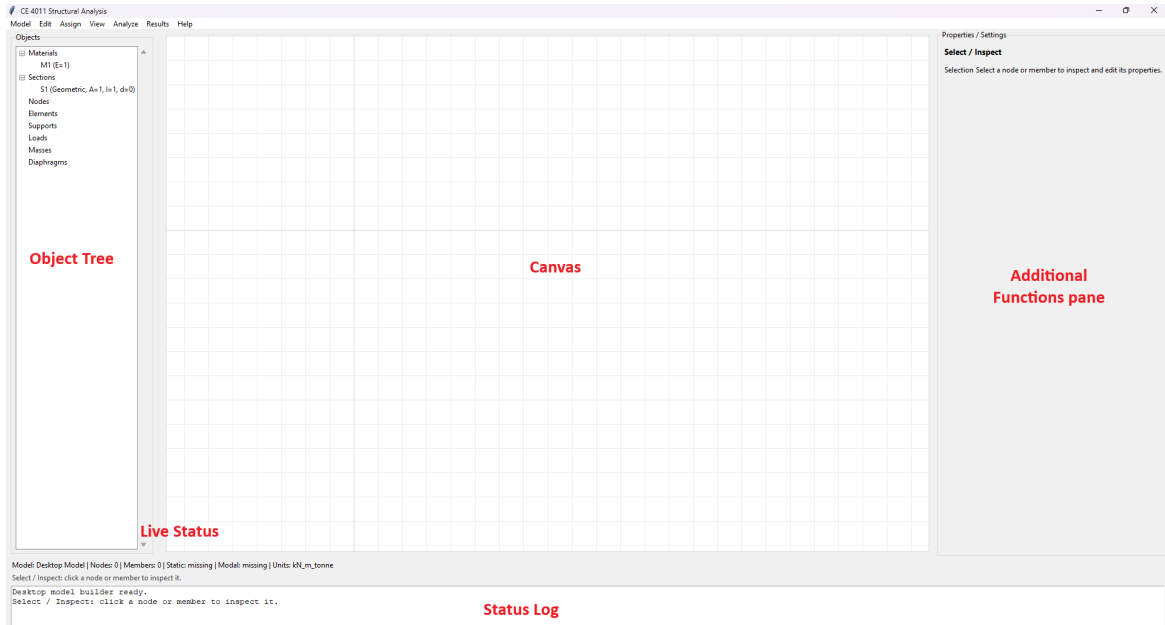


Figure 1: Main desktop UI

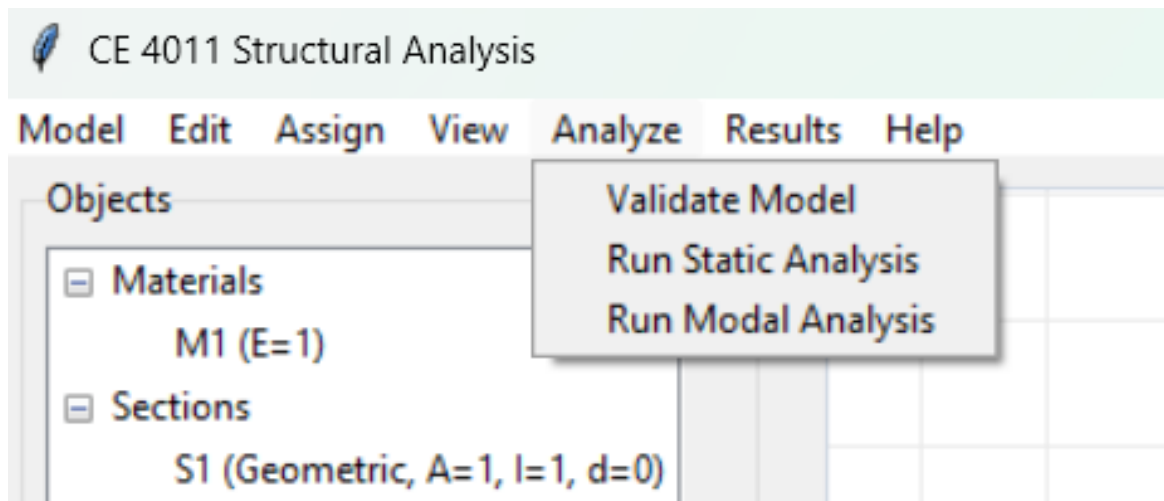


Figure 2: Model validation menu

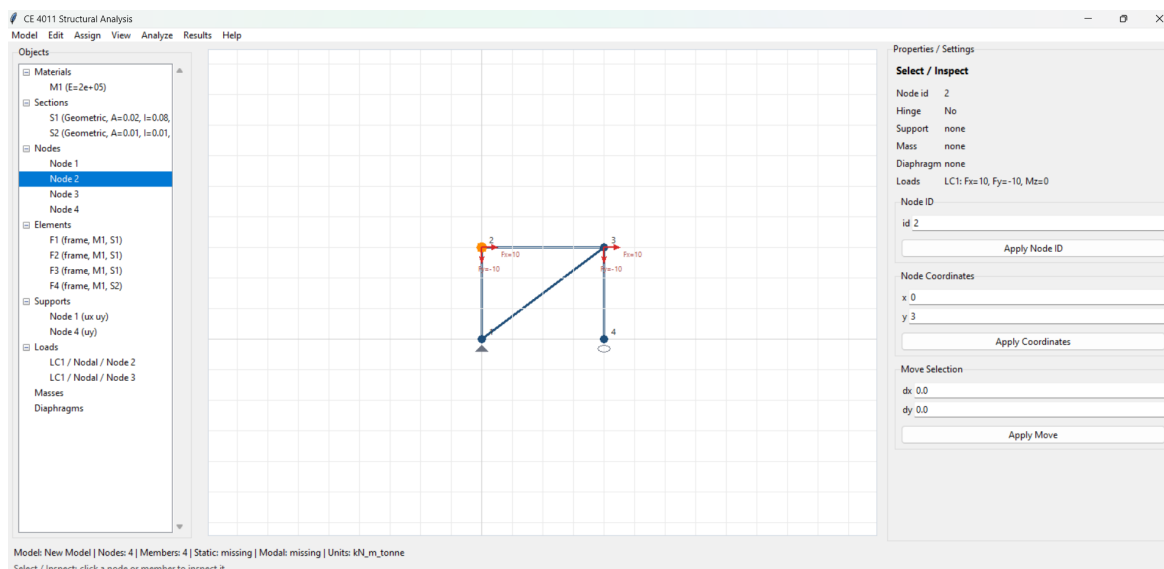


Figure 3: Property panel

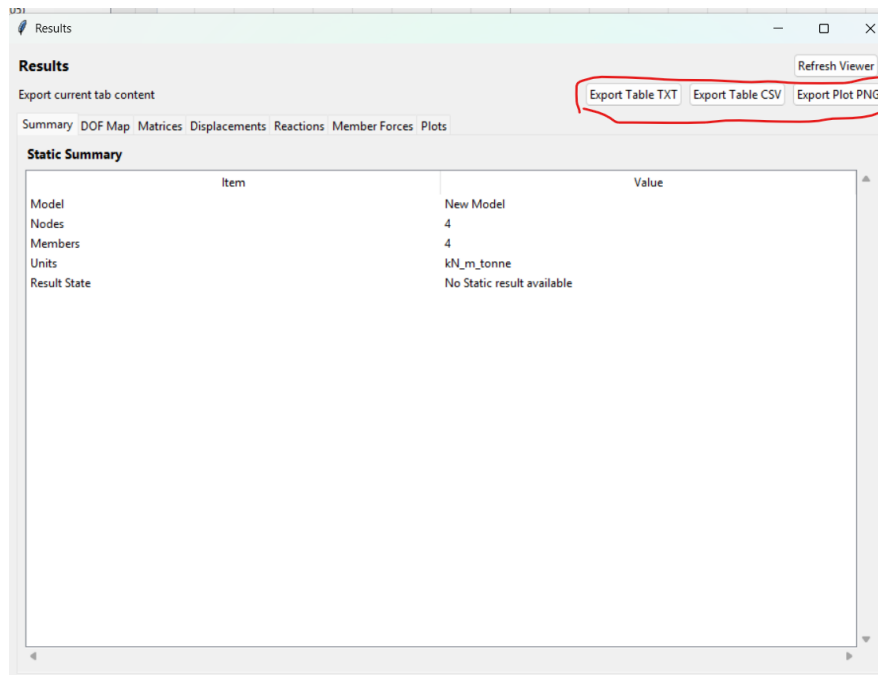


Figure 4: Export buttons

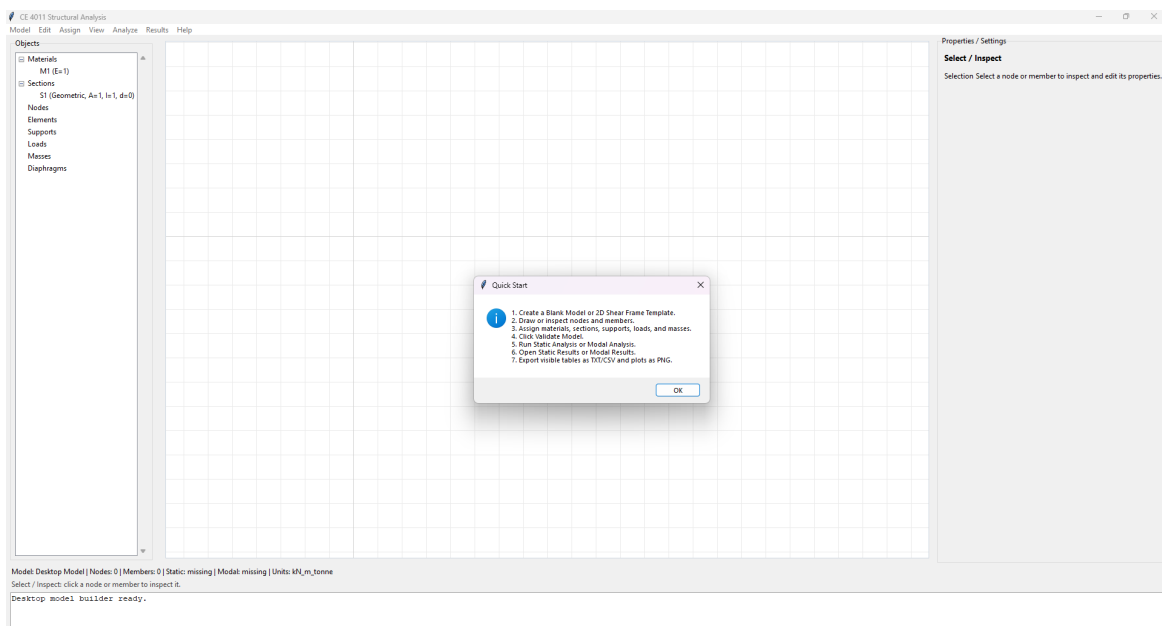


Figure 5: Quick-start workflow screen

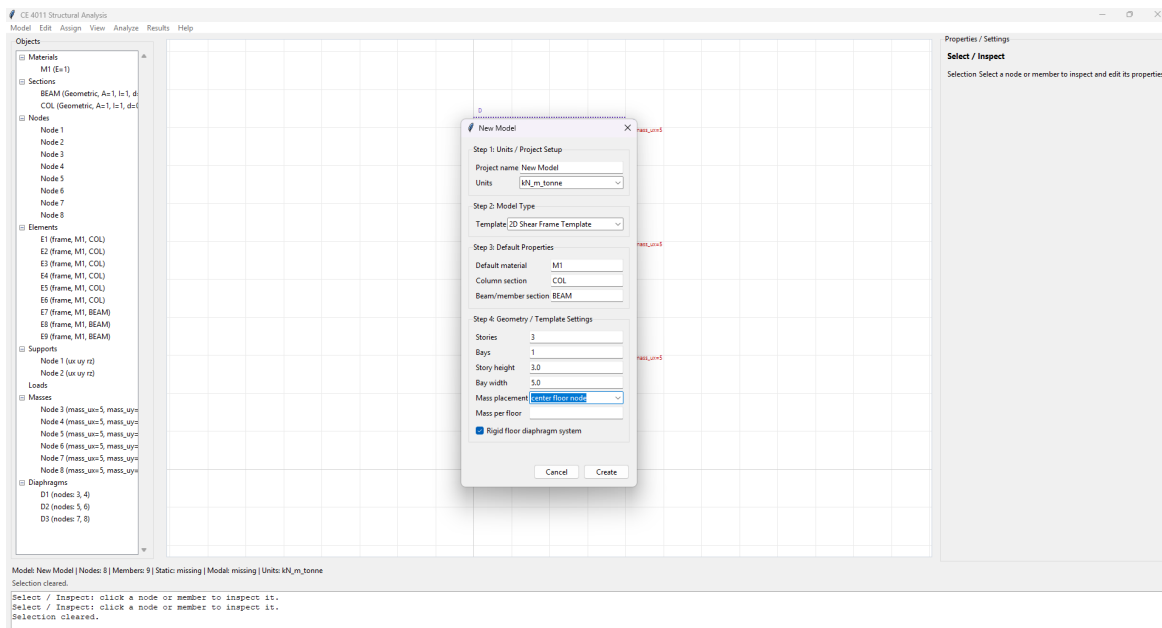


Figure 6: Shear-frame template input

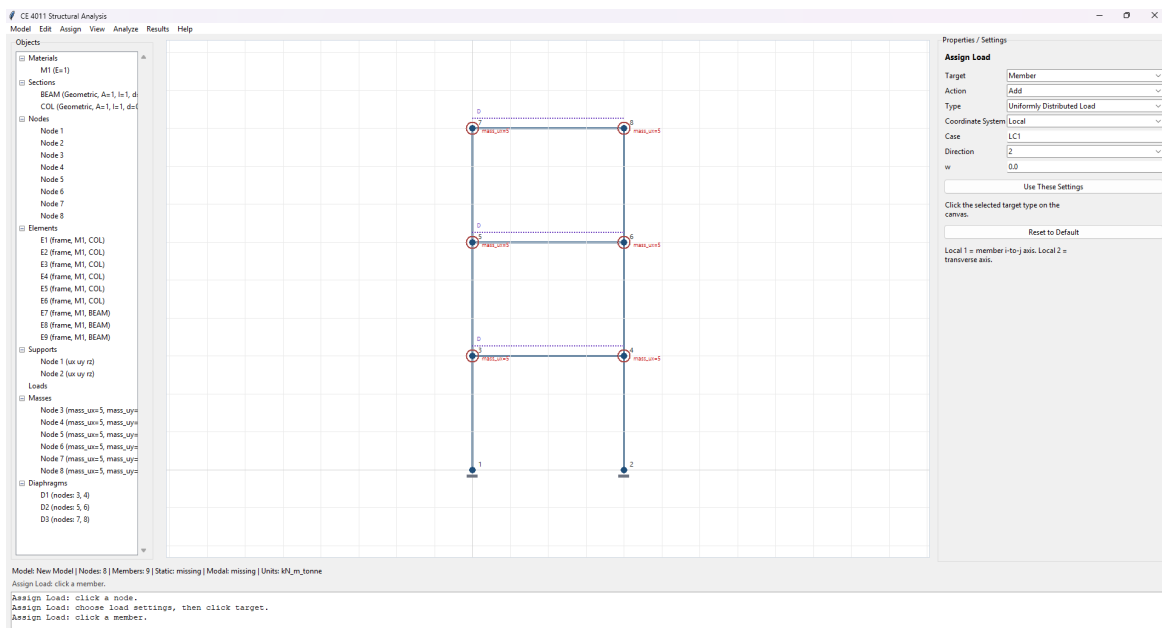


Figure 7: Member load assignment

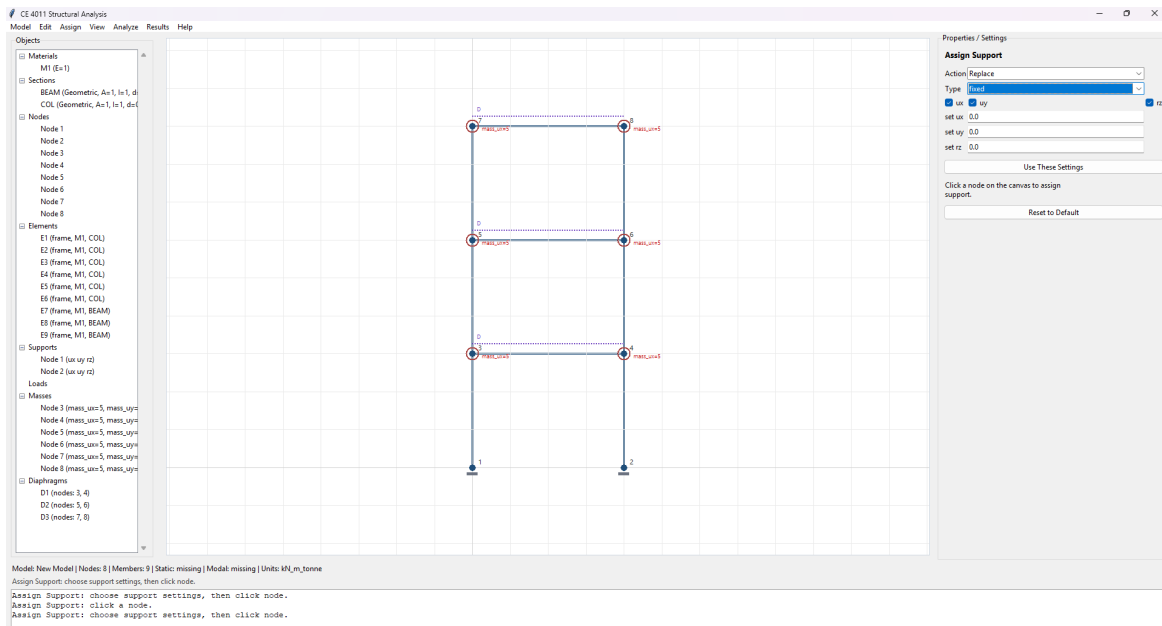


Figure 8: Support assignment

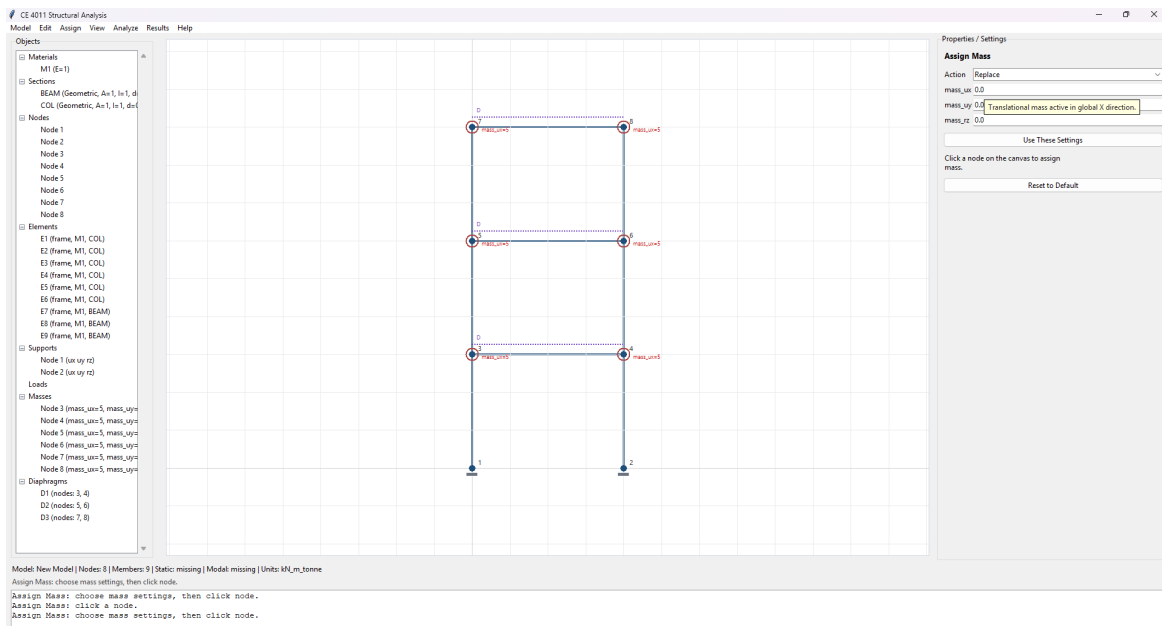


Figure 9: Mass assignment

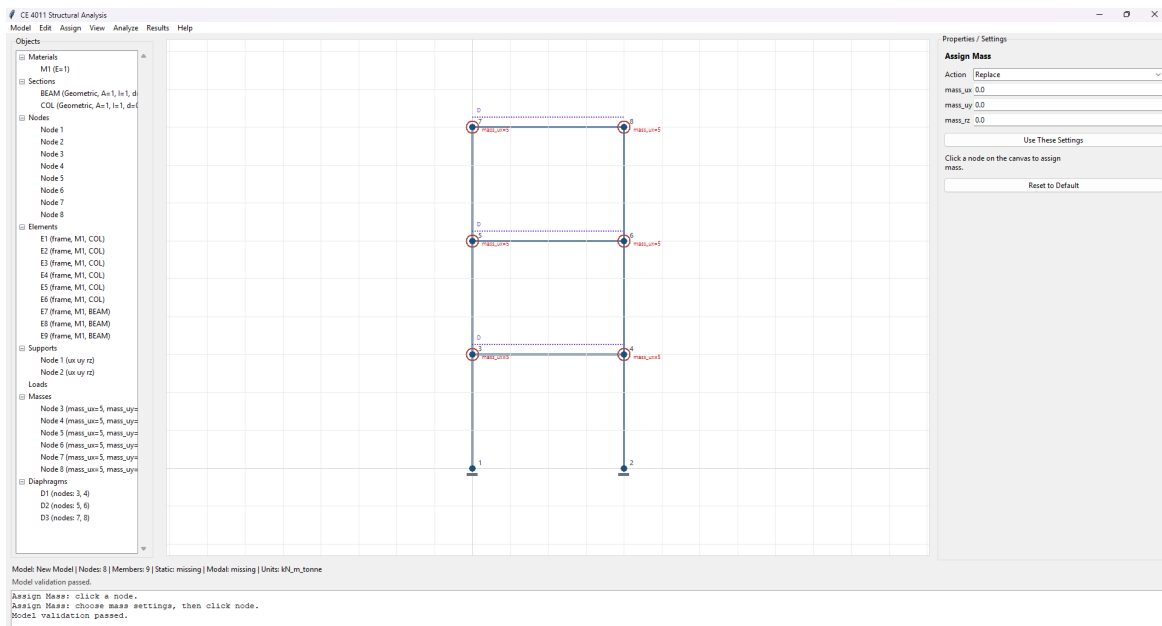


Figure 10: Successful model validation

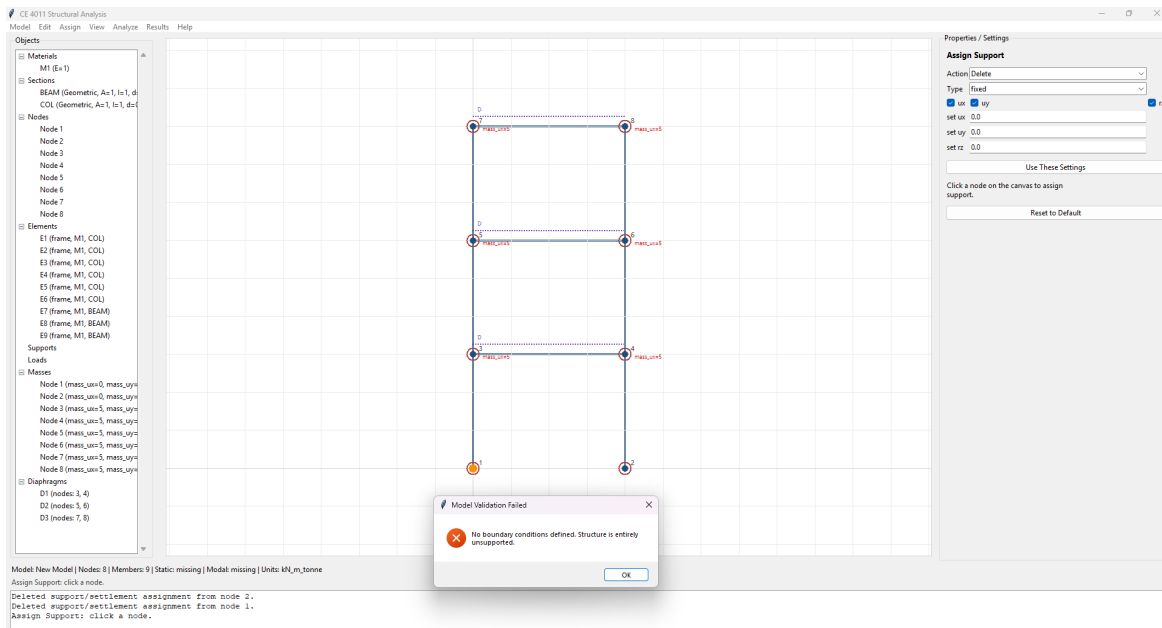


Figure 11: Unsuccessful model validation

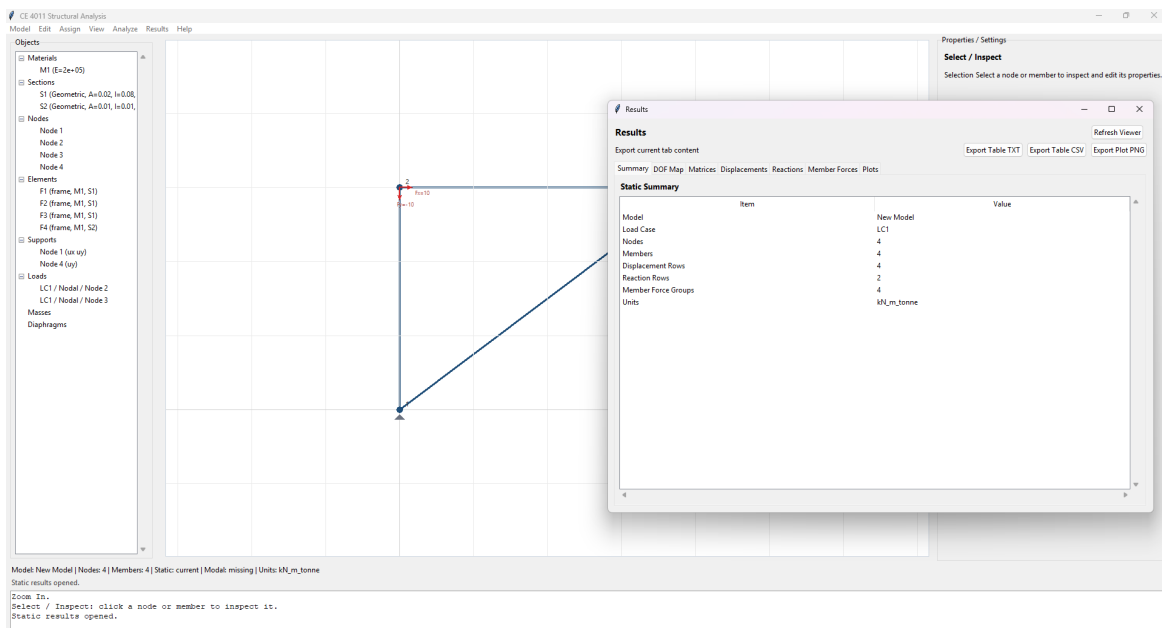


Figure 12: Static result summary

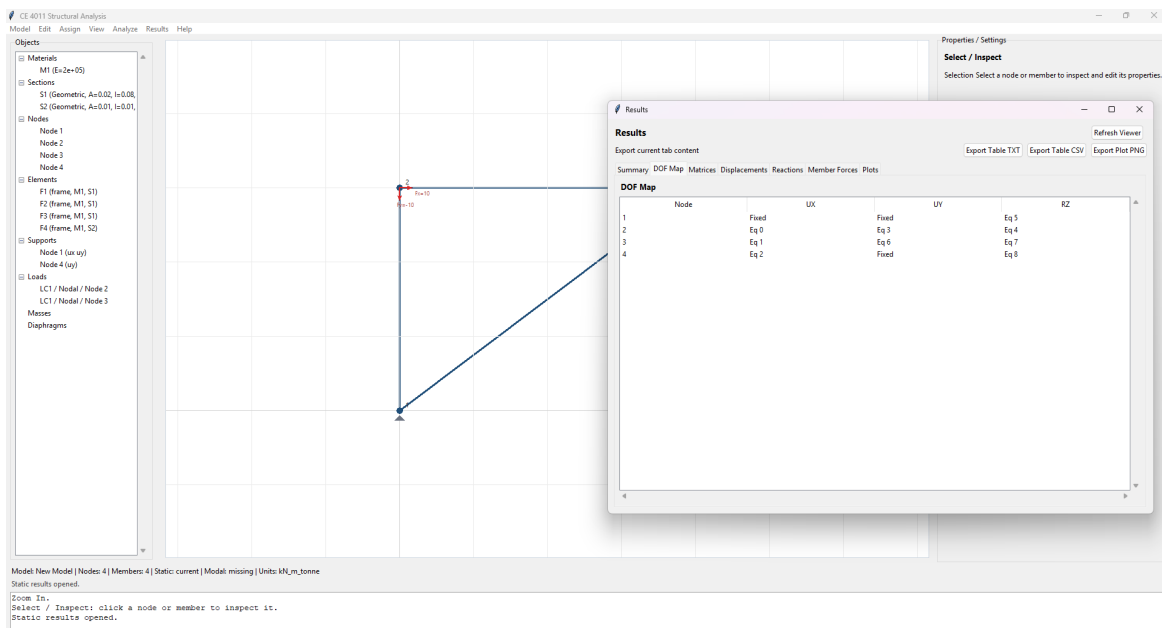


Figure 13: Static DOF map

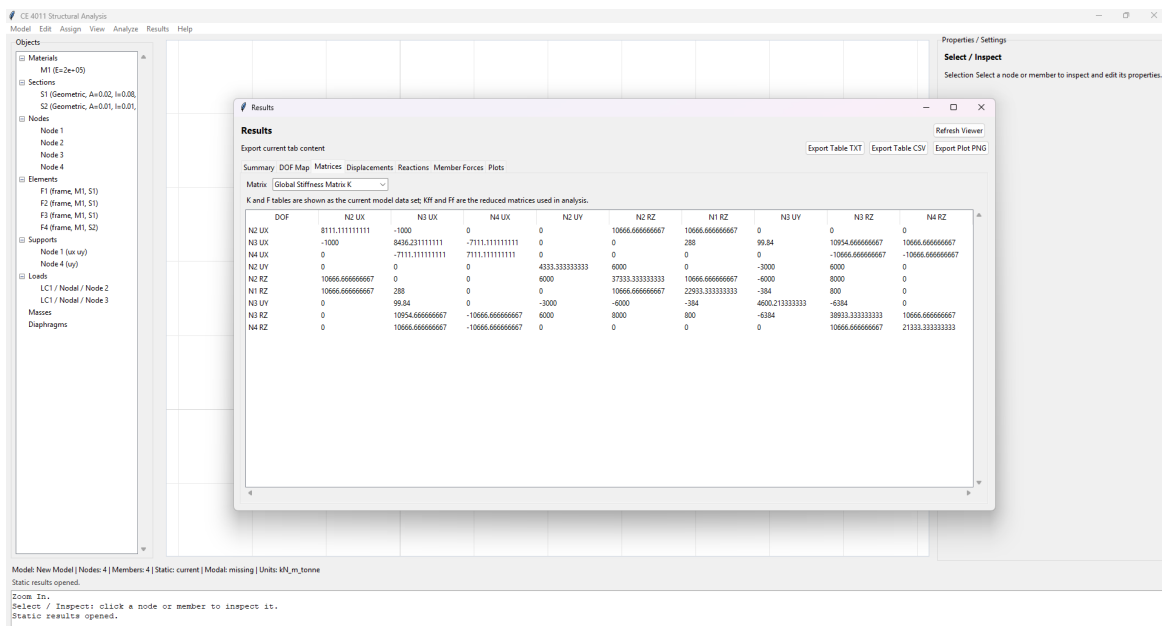


Figure 14: Static stiffness matrix

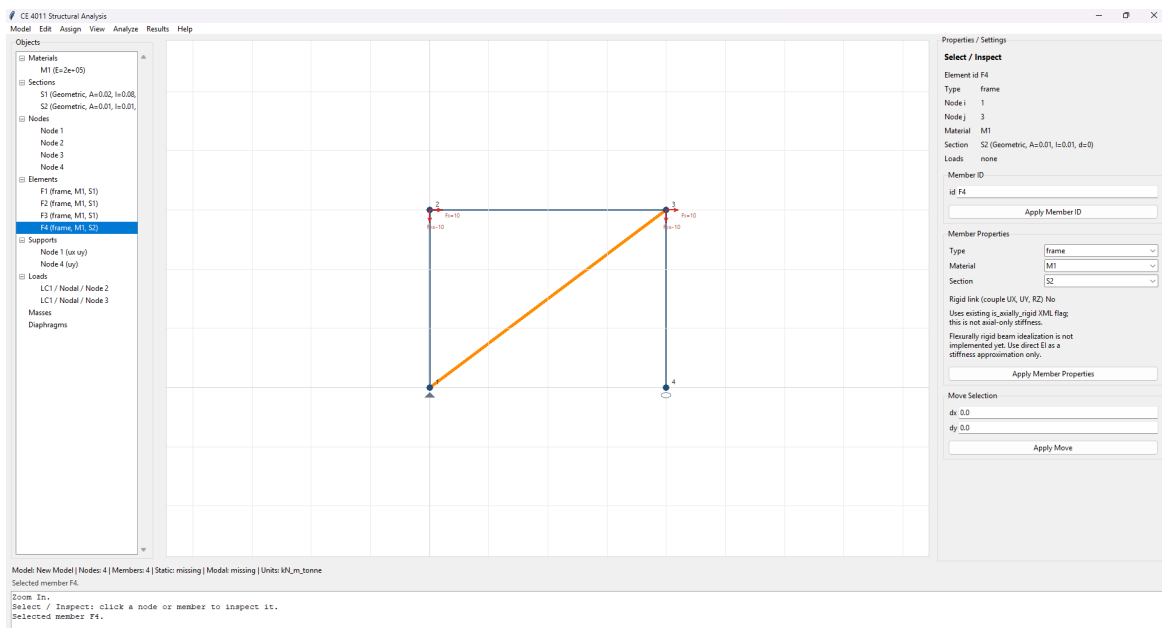


Figure 15: Static test-frame model

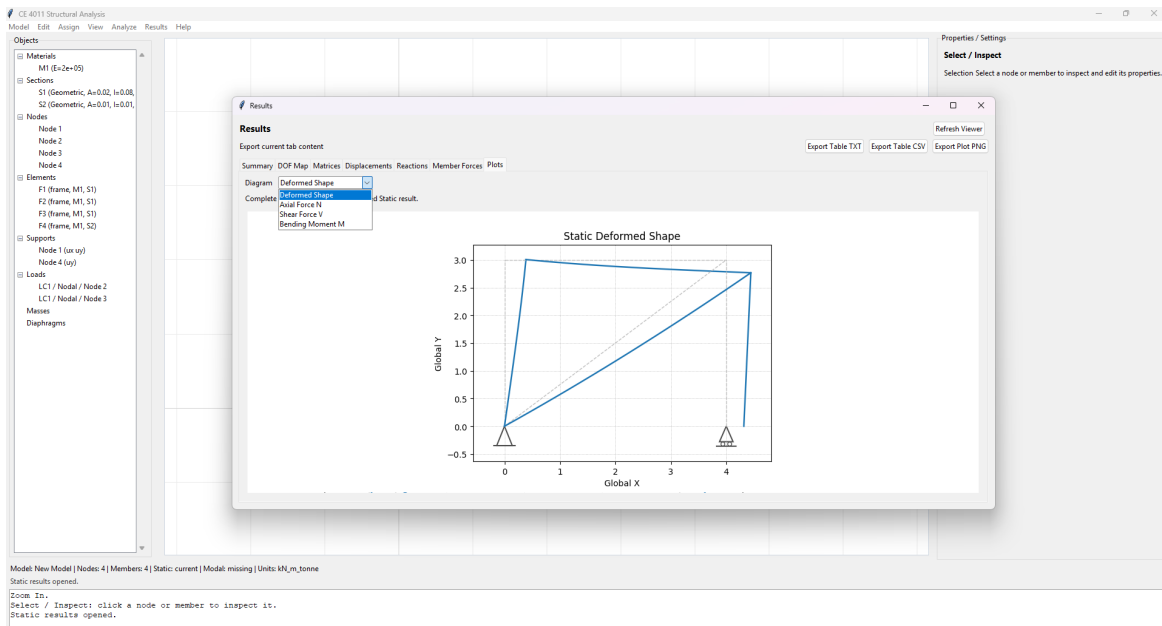


Figure 16: Static deformed shape

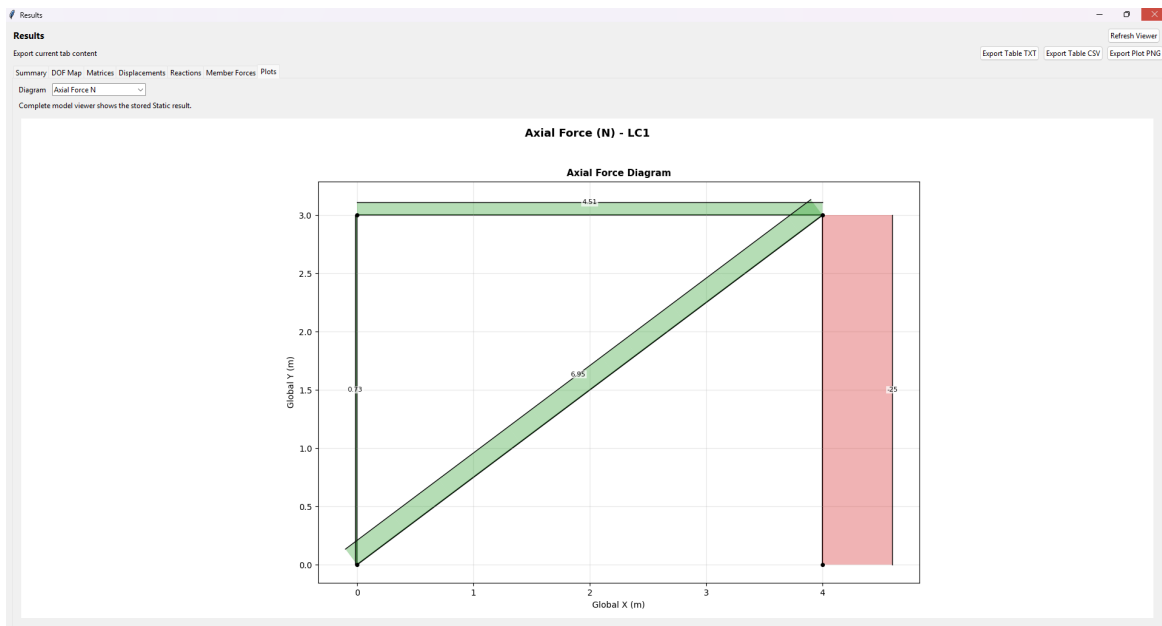


Figure 17: Static axial-force diagram

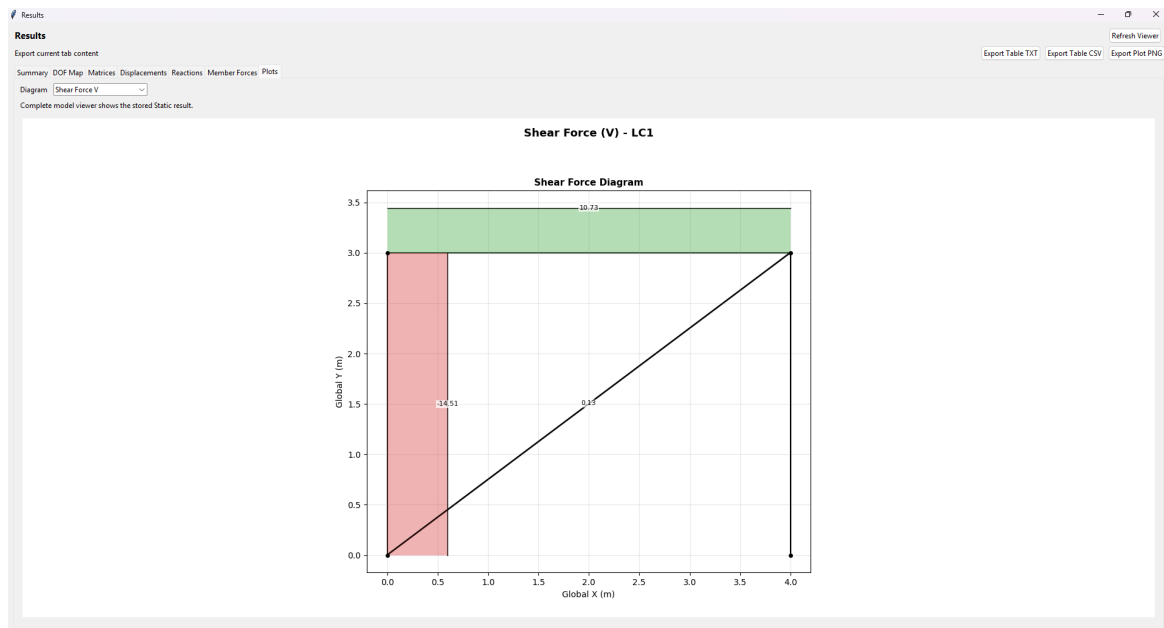


Figure 18: Static shear-force diagram

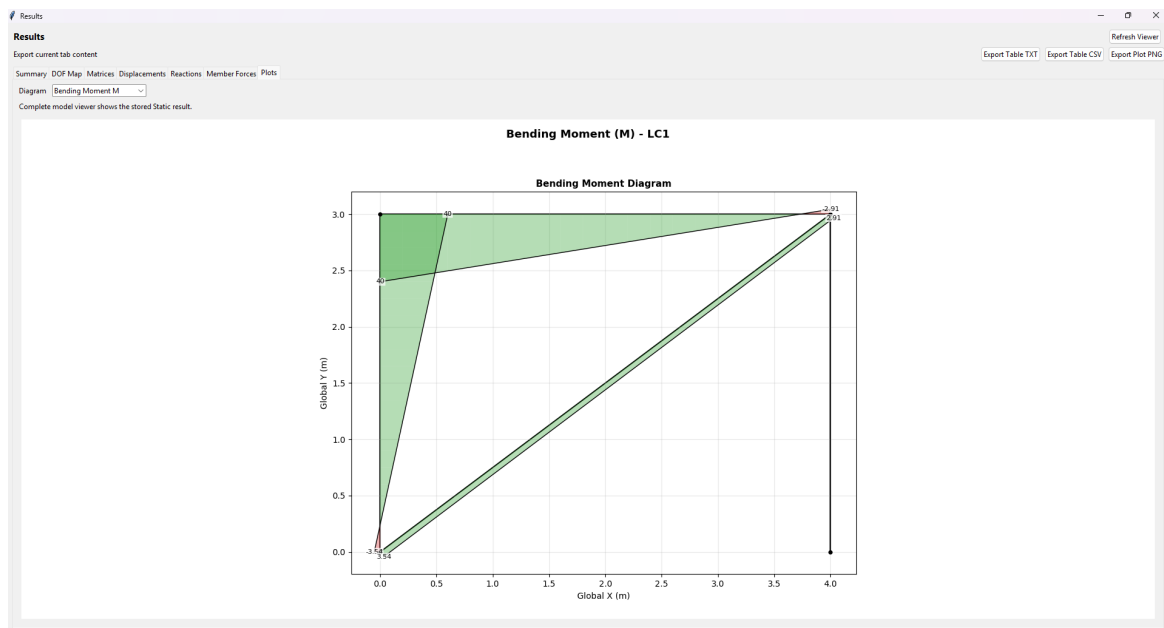


Figure 19: Static bending-moment diagram

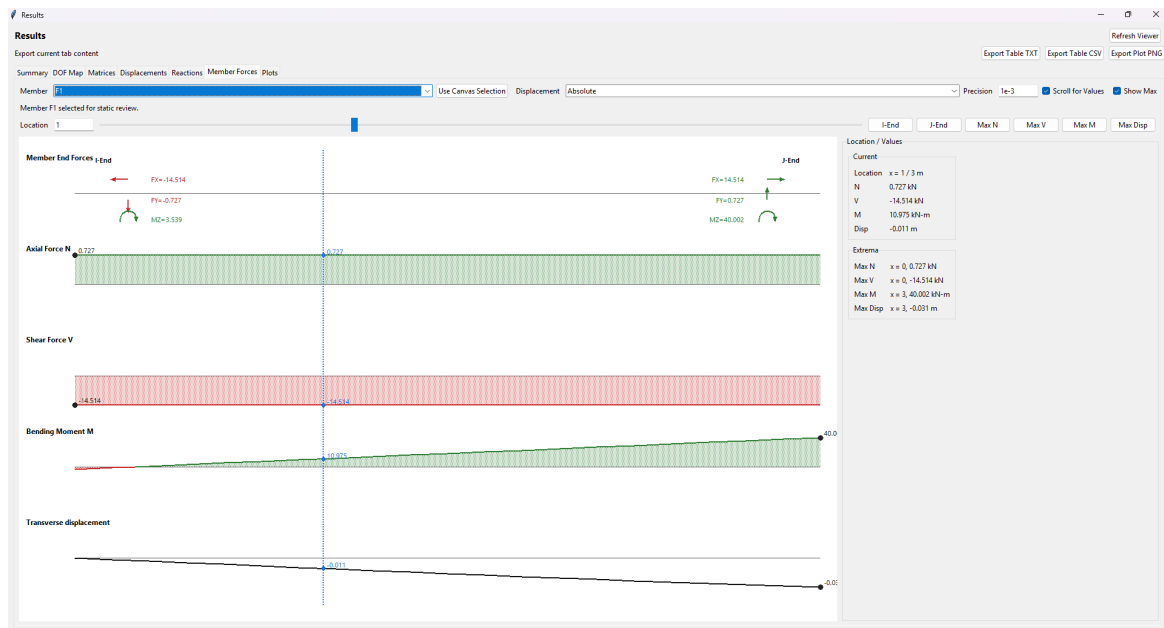


Figure 20: Individual member static result

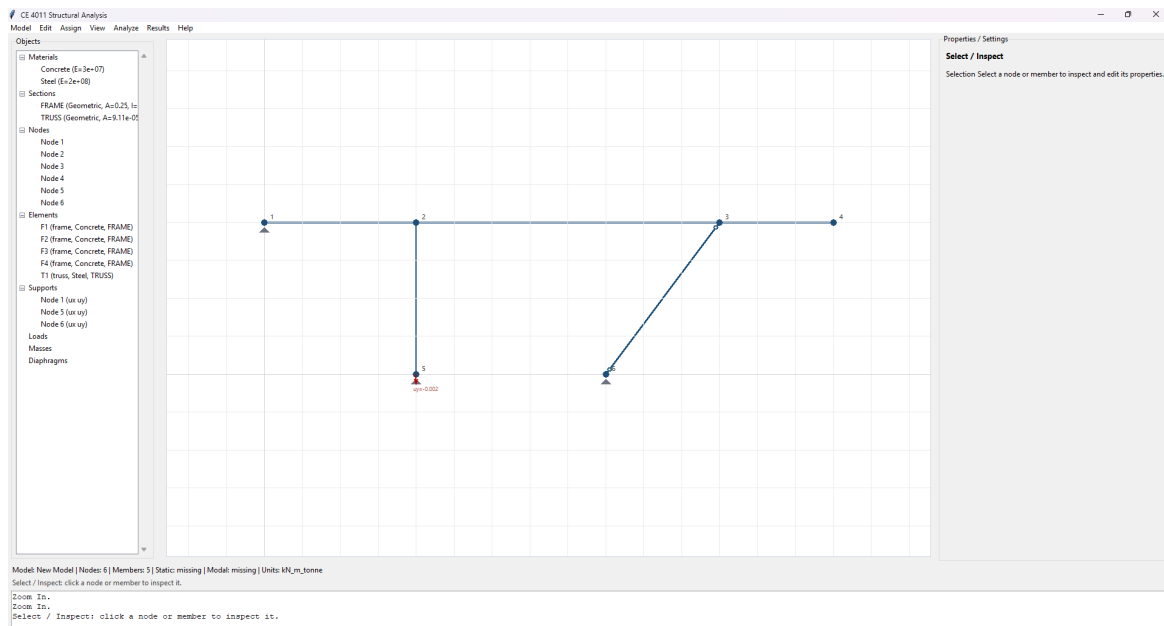


Figure 21: Settlement case screen

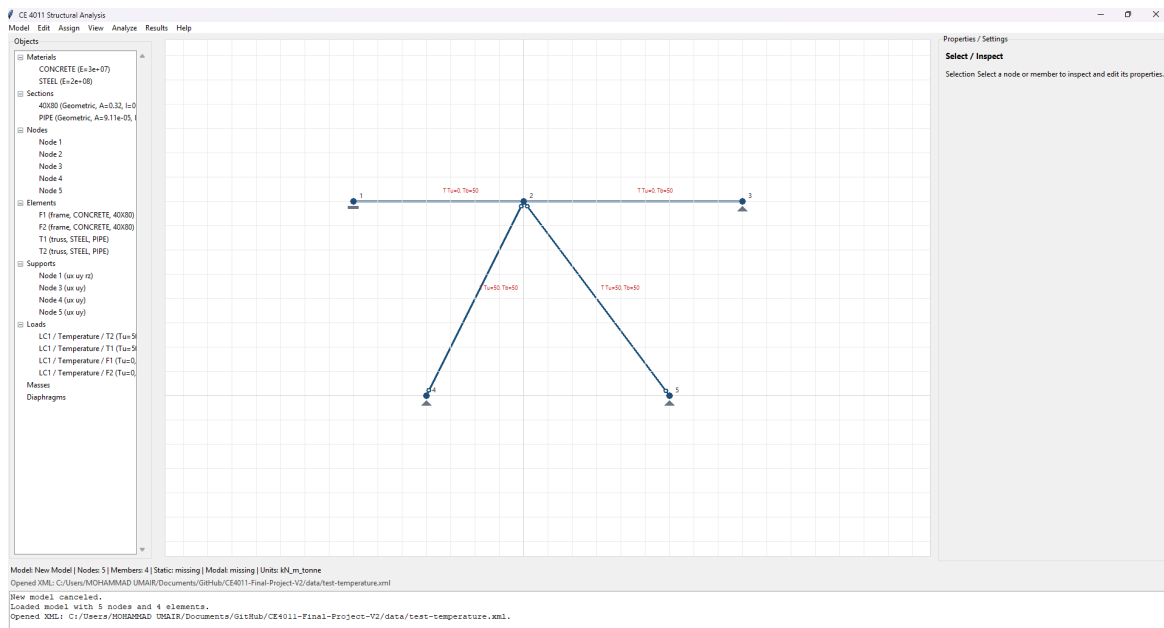


Figure 22: Temperature case screen

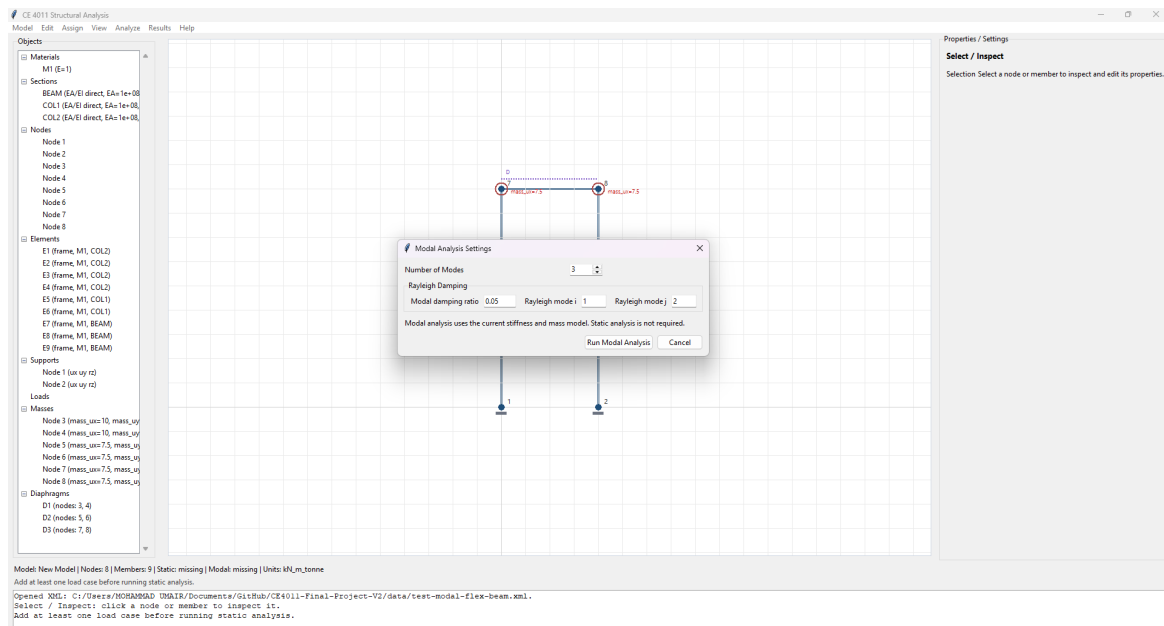


Figure 23: Modal settings

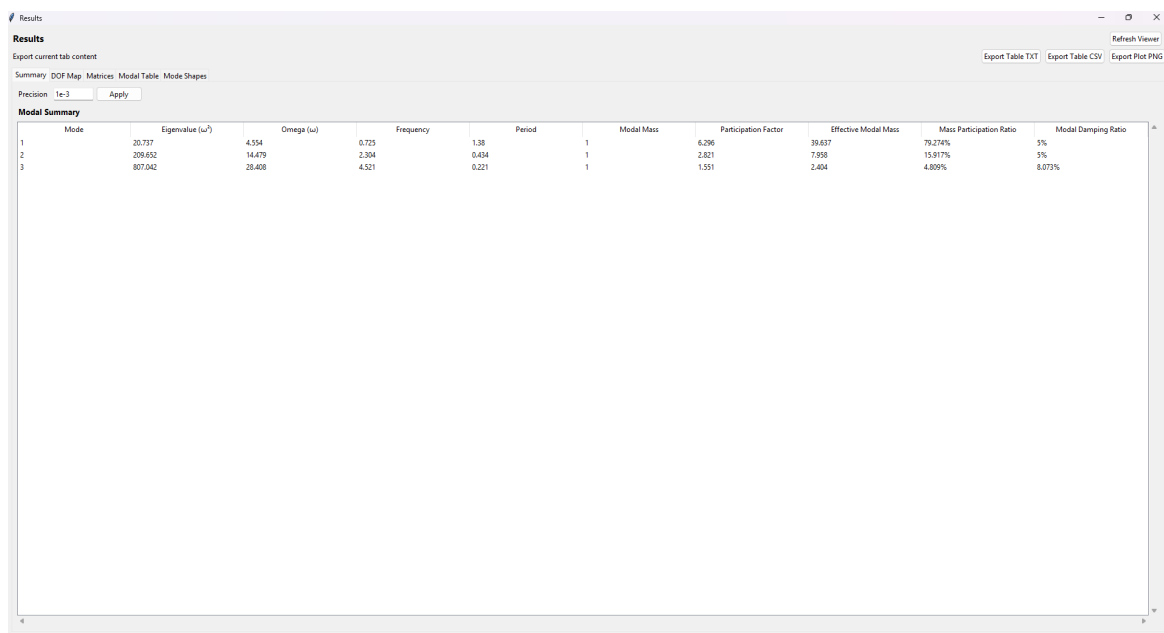


Figure 24: Modal result summary

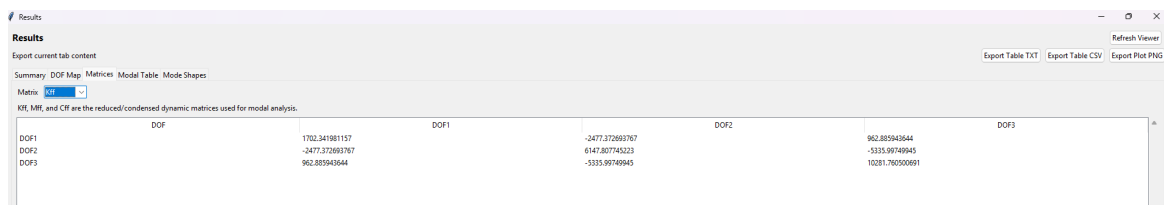


Figure 25: Modal stiffness matrix

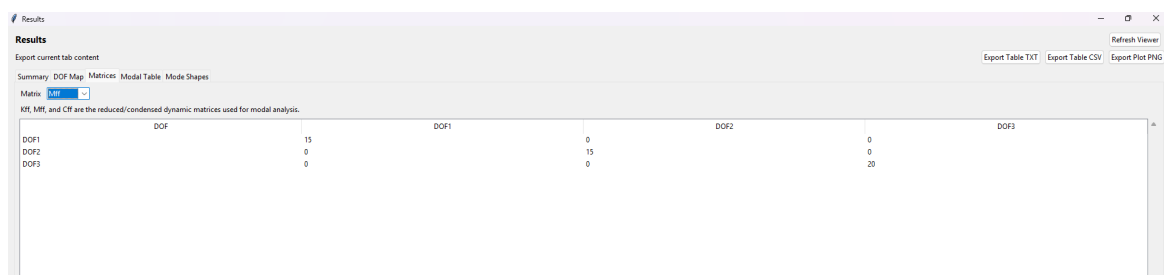


Figure 26: Modal mass matrix

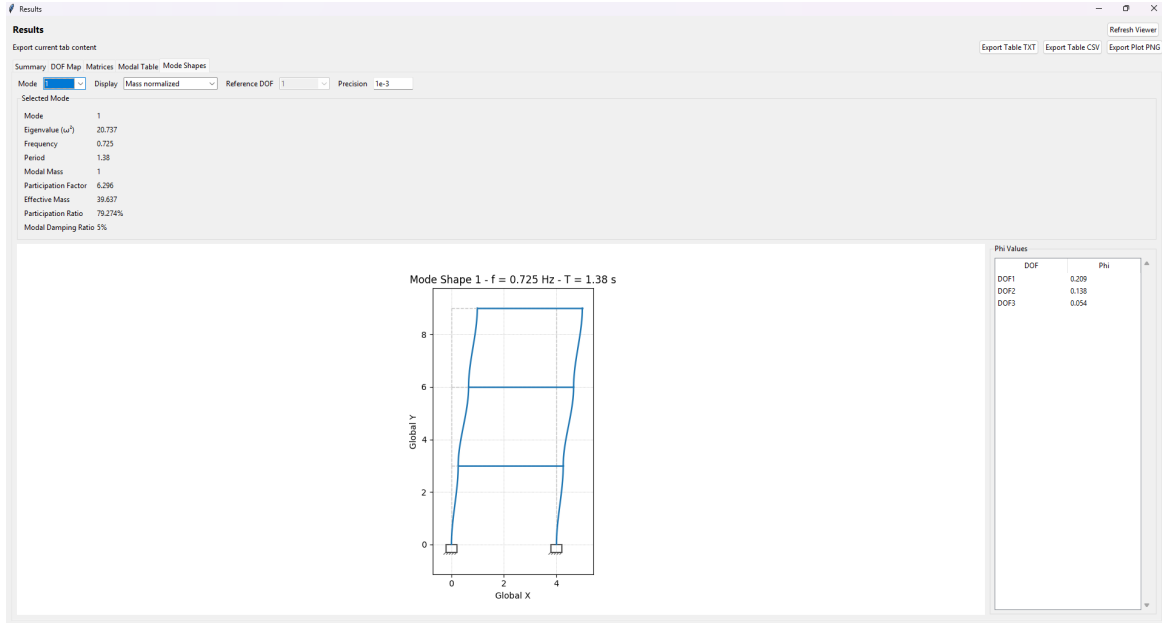


Figure 27: Mode-shape visualization

5 Static Frame Benchmark

The static frame benchmark verifies the direct stiffness workflow on a two-dimensional frame/truss-style model using retained XML input. The developed software path is:

XMLParser → DOFOptimizer → MatrixAssembler
→ BandedSolver → PostProcessor → StaticResults

The SAP2000 output is parsed into the project coordinate convention before comparison. This benchmark checks that the global stiffness assembly, boundary conditions, displacement solution, reaction recovery, and member-end force recovery remain consistent with the reference solution.

Table 2: Static result table export evidence from `data/test_frame_static_results_table.txt`

Exported field	Exported value
Model	New Model
Load case	LC1
Nodes and members	4 nodes; 4 members
Displacement rows	4
Reaction rows	2
Member force groups	4
Units	kN_m_tonne

The available static export is a visible-table summary, not a full numerical export of displacements, reactions, and member forces. The benchmark values below were recomputed from the retained XML through the same solver path used by the desktop application and compared with the retained SAP2000 reference text output after SAP2000 X-Z quantities were mapped to the project X-Y/ R_z convention.

Table 3: Static frame benchmark comparison table

Quantity	Developed software result	Reference result	Percent difference	Tolerance	Pass/fail
Joint 2 displacement u_x	3.1277175×10^{-2} m	3.1277×10^{-2} m	0.0006%	2% disp.	Pass
Joint 2 displacement u_y	5.4510749×10^{-4} m	5.4500×10^{-4} m	0.0197%	2% disp.	Pass
Joint 2 rotation r_z	$-8.0362030 \times 10^{-3}$ rad	-8.0360×10^{-3} rad	0.0025%	5% rot.	Pass
Joint 1 support reaction F_x	-20.0000 kN	-20.0000 kN	0.0000%	2% force	Pass
Joint 1 support reaction F_y	-5.0000 kN	-5.0000 kN	0.0000%	2% force	Pass
Joint 4 support reaction F_y	25.0000 kN	25.0000 kN	0.0000%	2% force	Pass
Frame F1 J-end moment M_z	40.0018 kN-m	40.0018 kN-m	0.0001%	2% moment	Pass
Frame F2 I-end shear F_y	-10.7268 kN	-10.7270 kN	0.0018%	2% force	Pass
N/V/M diagram endpoint, F1 $M(x = L)$	40.0018 kN-m	40.0018 kN-m	0.0001%	2% moment	Pass

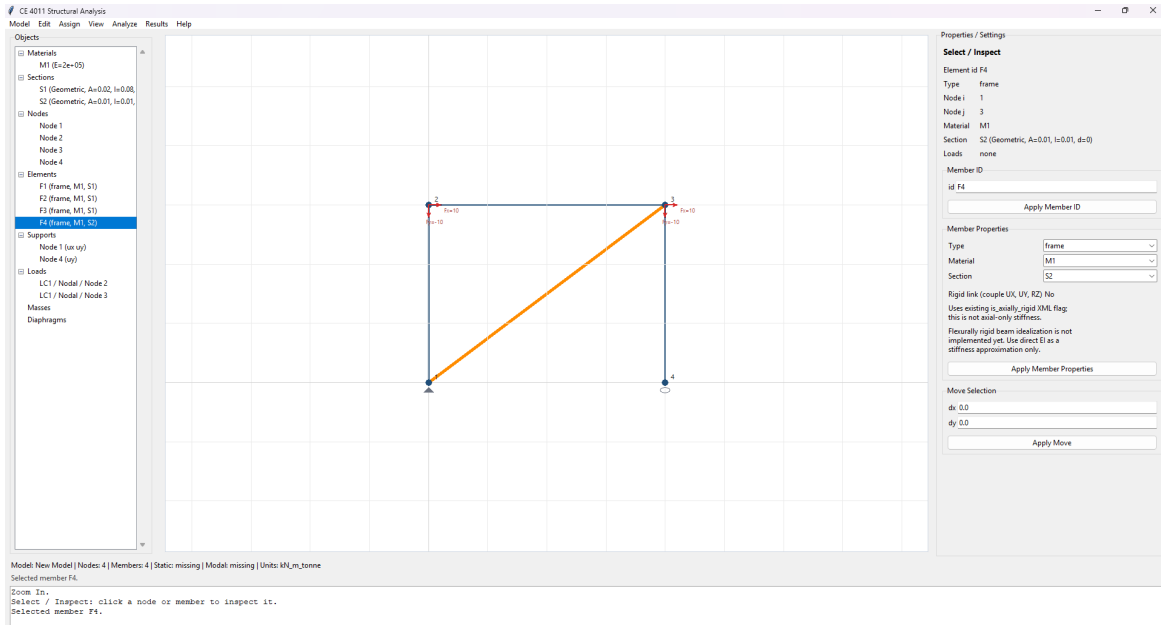


Figure 28: Static frame benchmark screen

6 Settlement Benchmark

The settlement benchmark verifies prescribed support displacement handling. For restrained displacement vector u_r , the reduced system solved by the static workflow is

$$K_{ff}u_f = F_f - K_{fr}u_r.$$

After the full displacement vector is reconstructed, reactions are recovered from $R = Ku - F$. The retained benchmark includes a vertical settlement at a restrained support and compares the resulting displacement and reaction fields with SAP2000 output.

Table 4: Settlement benchmark comparison table

Quantity	Developed software result	Reference result	Percent difference	Tolerance	Pass/fail
Prescribed settlement value used in assembly	XML node 5: $u_y = -0.002$ m	Same XML input	0.000%	Exact input check	Pass
Joint 3 displacement u_x	5.3686×10^{-6} m	5.3200×10^{-6} m	0.9129%	1% disp.	Pass
Joint 3 displacement u_y	-9.5686×10^{-4} m	-9.5400×10^{-4} m	0.3003%	1% disp.	Pass
Joint 3 rotation r_z	2.8132×10^{-4} rad	2.8000×10^{-4} rad	0.4715%	5% rot.	Pass
Node 5 settled-support reaction F_x	5.0658 kN	4.9930 kN	1.4587%	2% force	Pass
Node 5 settled-support reaction F_y	-11.7328 kN	-11.6370 kN	0.8232%	2% force	Pass
Frame F4 I-end shear F_y	-11.7328 kN	-11.6370 kN	0.8232%	2% force	Pass
Frame F4 J-end moment M_z	-20.2633 kN-m	-19.9705 kN-m	1.4663%	2% moment	Pass
N/V/M diagram endpoint, F4 $M(x = L)$	-20.2633 kN-m	-19.9705 kN-m	1.4663%	2% moment	Pass

7 Temperature Benchmark

The temperature benchmark verifies thermal fixed-end effects and transformation-sensitive force recovery. The XML input contains uniform and combined temperature loads on frame and truss members. The inclined truss members are especially useful because errors in local/global transformation, thermal force sign, or SAP2000 coordinate mapping create visible force mismatches.

The strict thermal benchmark checks displacements, reactions, and member-end force components. The automated strict test uses a displacement tolerance of 1.0×10^{-5} , a force tolerance of 0.05, and a moment tolerance of 0.6 in the project units used by the benchmark.

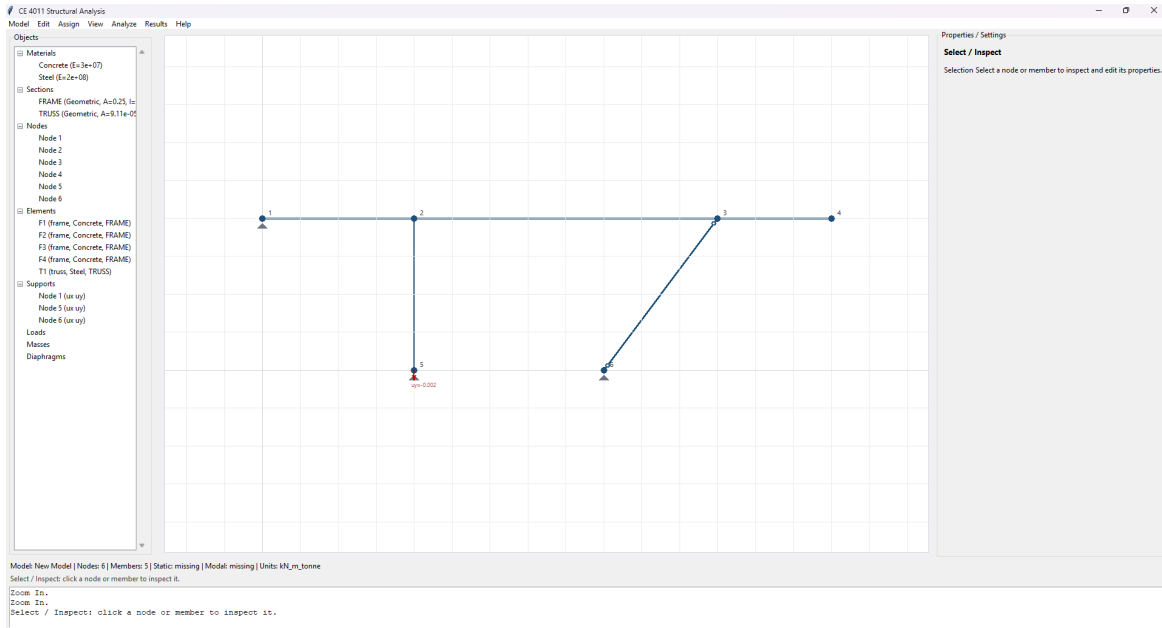


Figure 29: Settlement benchmark screen

Table 5: Temperature benchmark comparison table

Quantity	Developed software result	Reference result	Percent difference	Tolerance	Pass/fail
Joint 2 displacement u_y	-1.9385×10^{-3} m	-1.9300×10^{-3} m	0.4418%	1.0×10^{-5} abs	Pass
Joint 2 rotation r_z	-4.5932×10^{-4} rad	-4.5400×10^{-4} rad	1.1714%	1.0×10^{-5} abs	Pass
Joint 1 support reaction F_x	1921.0043 kN	1921.0040 kN	0.0000%	0.05 force	Pass
Joint 1 support reaction F_y	5.9275 kN	5.9110 kN	0.2794%	0.05 force	Pass
Joint 1 support moment M_z	310.3421 kN-m	309.8603 kN-m	0.1555%	0.6 moment	Pass
Frame F2 I-end moment M_z	268.8495 kN-m	268.4813 kN-m	0.1371%	0.6 moment	Pass
Inclined truss T2 J-end force F_x	-8.2546 kN	-8.2470 kN	0.0926%	0.05 force	Pass
Inclined truss T2 J-end force F_y	11.0062 kN	10.9960 kN	0.0926%	0.05 force	Pass
Frame F1 thermal end moment M_z	310.3421 kN-m	309.8603 kN-m	0.1555%	0.6 moment	Pass

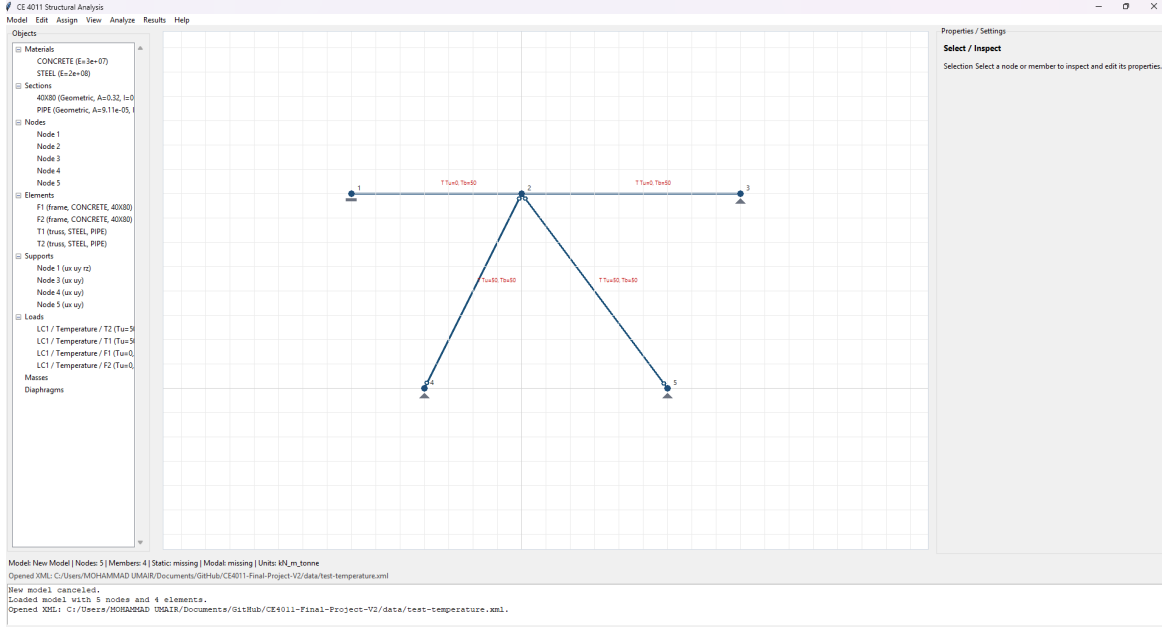


Figure 30: Temperature benchmark screen

8 Modal Flexible-Beam / Shear-Frame Benchmark

The modal benchmark verifies that the dynamic workflow preserves stiffness coupling while avoiding invalid massless dynamic DOFs. The benchmark uses a flexible-beam/shear-frame-style model, assembles the stiffness and lumped mass matrices, condenses stiffness-coupled massless DOFs where required, and solves

$$K\phi = \lambda M\phi.$$

The comparison checks the reduced or condensed stiffness matrix, eigenvalues, circular frequencies, periods, and normalized mode shapes against the stored reference solution.

Table 6: Modal result export values from `data/modal_results_table.csv`

Mode	λ	ω	f	T	Modal mass	Γ	Effective mass	Mass ratio
1	20.737	4.554	0.725	1.380	1.000	6.296	39.637	79.274%
2	209.652	14.479	2.304	0.434	1.000	2.821	7.958	15.917%
3	807.042	28.408	4.521	0.221	1.000	1.551	2.404	4.809%

Table 7: Modal flexible-beam benchmark comparison table

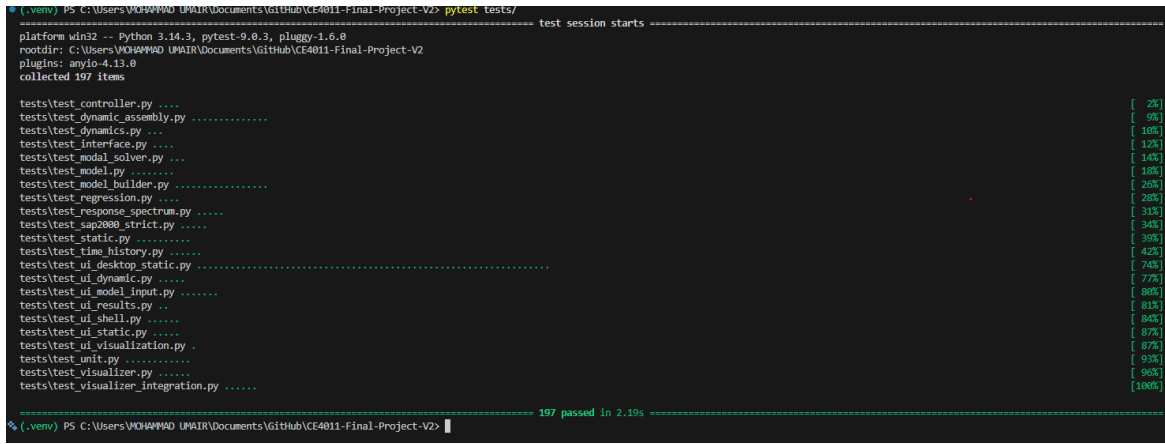
Quantity	Developed software result	Reference result	Percent difference	Tolerance	Pass/fail
Eigenvalue λ_1	20.737	20.739	0.010%	0.02 absolute	Pass
Circular frequency ω_1	4.554	4.554	0.000%	0.002 absolute	Pass
Period T_1	1.380	1.380	0.000%	0.002 absolute	Pass

9 Automated Pytest Validation

The final stable testing baseline is:

```
pytest tests/ -q -> 197 passed
pytest tests/test_ui_desktop_static.py -q -> 63 passed
pytest tests/test_ui_desktop_static.py
      tests/test_visualizer.py tests/test_visualizer_integration.py -q
      -> 76 passed
```

The final hardening changed only `tests/test_ui_desktop_static.py`. No solver math, XML schema, fixtures, or source behavior changed during final testing hardening. Static benchmark tests now use retained `data/test-frame.xml`. The file `data/ground_motion.txt` was restored only for legacy/future THA backend compatibility and does not change the final desktop scope.



```
(.venv) PS C:\Users\UMAIR\Documents\GitHub\CE4811-Final-Project-V2> pytest tests/
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\UMAIR\Documents\GitHub\CE4811-Final-Project-V2
plugins: anyio-4.13.0
collected 197 items

tests/test_controller.py ..... [ 25%]
tests/test_dynamic_assembly.py ..... [ 95%]
tests/test_dynamics.py ... [ 105%]
tests/test_interface.py ..... [ 125%]
tests/test_modal_solver.py ... [ 145%]
tests/test_model.py ..... [ 165%]
tests/test_model_builder.py ..... [ 205%]
tests/test_regression.py ..... [ 225%]
tests/test_response_spectrum.py ..... [ 245%]
tests/test_sap2000_strict.py ..... [ 315%]
tests/test_static.py ..... [ 345%]
tests/test_time_history.py ..... [ 395%]
tests/test_ui_desktop_static.py ..... [ 745%]
tests/test_ui_dynamic.py ..... [ 775%]
tests/test_ui_model_input.py ..... [ 805%]
tests/test_ui_results.py .. [ 815%]
tests/test_ui_shell.py ..... [ 845%]
tests/test_ui_static.py ..... [ 875%]
tests/test_ui_visualization.py . [ 875%]
tests/test_unit.py ..... [ 935%]
tests/test_visualizer.py ..... [ 965%]
tests/test_visualizer_integration.py ..... [ 100%]

===== 197 passed in 2.19s =====
(.venv) PS C:\Users\UMAIR\Documents\GitHub\CE4811-Final-Project-V2>
```

Figure 31: Test suite output

Table 8: Automated test coverage summary

Test area	Representative tests	Checked behavior	Result
XML parsing/export	XML parser/export tests and desktop XML command-path test	Save/load/export reproducibility and parse-back through XMLParser	Passed
Model building	ModelBuilder tests and template workflows	Programmatic, template, table, and XML-backed model creation	Passed
DOF and stiffness assembly	Static solver and assembly tests	DOF maps, element stiffness, global K , reduced K_{ff} , and force vectors	Passed
Static result recovery	Static, post-processing, and benchmark tests	Displacements, reactions, member forces, member-end force signs, and N/V/M diagram data	Passed
Settlement and temperature behavior	Interface, unit, strict SAP2000, and regression tests	Prescribed support displacement assembly, thermal fixed-end effects, and transformation-sensitive force recovery	Passed
Dynamic/modal assembly	Modal solver, dynamic assembly, and regression tests	Mass assembly, active dynamic DOFs, massless-DOF handling, eigenvalues, frequencies, and periods	Passed

Test area	Representative tests	Checked behavior	Result
Modal properties	Modal solver and desktop modal tests	Mass normalization, participation factors, effective modal mass, participation ratios, and display normalization options	Passed
Visualization	Visualizer and visualizer integration tests	Model preview, deformed shape, diagrams, mode-shape plotting adapters, and missing-field tolerance	Passed
Desktop workflow/export	UI desktop static tests	Static table TXT export, Modal table CSV export, Static plot PNG export, XML save/export parse-back, stale-result clearing, and final Static/Modal-only desktop scope	Passed
Reference validation	SAP2000/reference comparison tests	Agreement with stored reference solutions within the tolerances described above	Passed

10 Desktop XML/Export Command-Path Validation

Two final P0 desktop workflow tests were added during final hardening:

1. A headless XML command-path test for `_save_xml()` and `_export_xml()`, with the written XML parsed back through `XMLParser`.
2. A result export test for Static table TXT export, Modal table CSV export, and Static plot PNG export through the shared export path.

These tests address the previous main risk area: desktop workflow coverage rather than numerical correctness. They confirm that the final UI command paths reach the XML and result-export backends without requiring a manual GUI session.

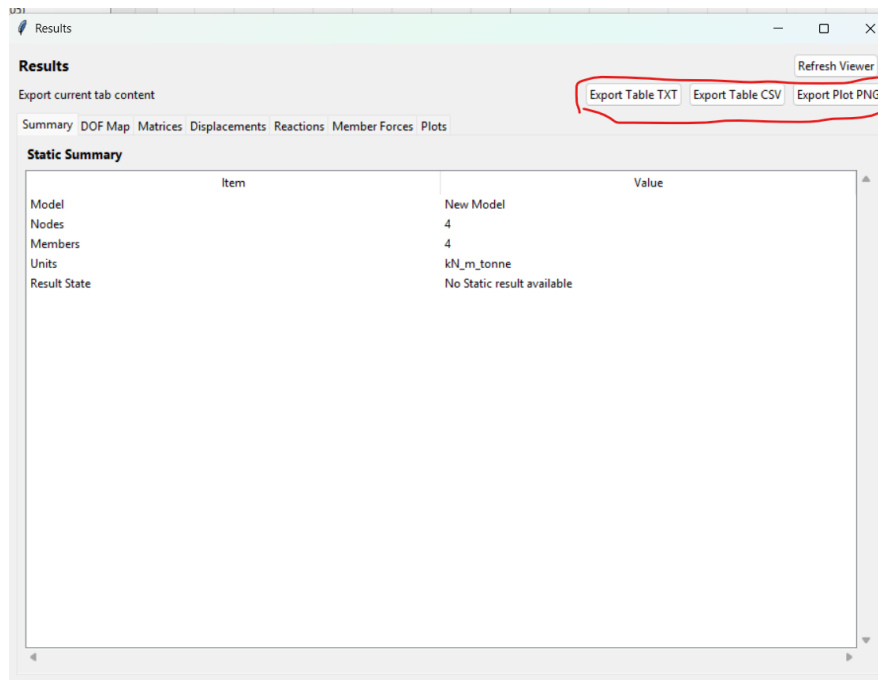


Figure 32: Export buttons validation screen

Table 9: Desktop export evidence from retained `data/` outputs

Export artifact	Export type	Verified evidence in retained file
Static table	Static TXT table	Source <code>data/test_frame_static_results_table.txt</code> : model <code>New Model</code> ; load case <code>LC1</code> ; 4 displacement rows; 2 reaction rows; 4 member-force groups; <code>kN_m_tonne</code> units.
Static plots	PNG figure evidence	Static plot figures are already placed under <code>docs/final_latex/main_report/figures/</code> : deformed shape, axial force, shear force, and bending moment. No standalone static plot export file is present under <code>data/</code> .
Modal table	Modal CSV table	Source <code>data/modal_results_table.csv</code> : three modal rows with eigenvalues, frequencies, periods, modal mass, participation factors, effective modal mass, participation ratios, and damping ratios.
Modal plots	Modal PNG plots	Source <code>data/modal_results_plot*.png</code> : three exported modal plot PNG files exist in <code>data/</code> ; LaTeX figure references remain limited to images already placed under <code>docs/final_latex/</code> .